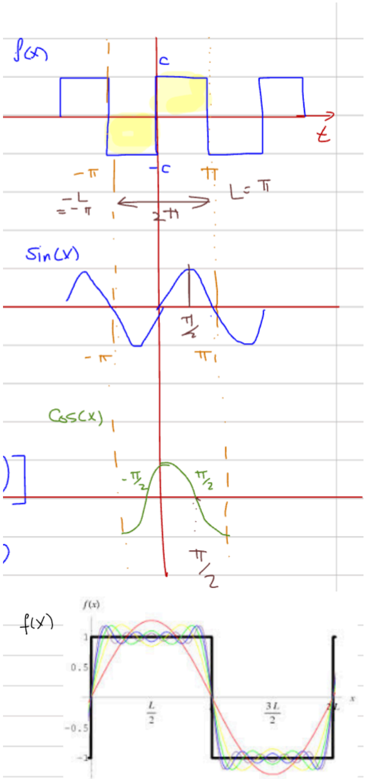
4 - Physical Layer Notes

Fourier Series Representation



The Fourier series is used to represent signals in the physical layer. The general representation of a function $f(x)$ is defined as:

- $f(x) = a_0 + \sum_{n=1}^{N}[a_n \cos(\frac{n\pi}{L}x) + b_n \sin(\frac{n\pi}{L}x)]$

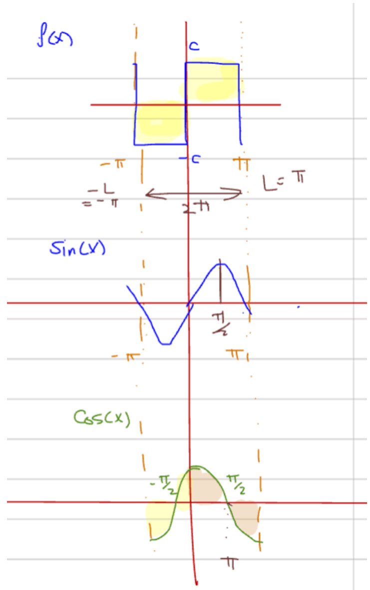
When the period is $P = 2L$, the coefficients are calculated as follows:

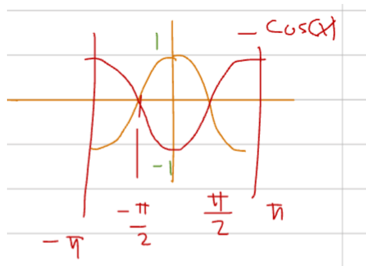
- $a_0 = \frac{1}{2L}\int_{-L}^{L} f(x)dx$
- $a_n = (\frac{1}{L})\int_{-L}^{L} f(x)\cos(\frac{n\pi}{L}x)dx$
- $b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin(\frac{n\pi}{L}x)dx$

Square Wave Function and Fourier Approximation

Example 1: Square Wave Function

Problem: Find the Fourier series coefficients for a square wave function with period $P = 2\pi$ (where $L = \pi$).





Answer:

Finding a_1:

$$a_1 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)\cos(x)dx \qquad = \frac{1}{\pi}\left[\int_{-\pi}^{0} (-c)\cos(x)dx + \int_{0}^{\pi} c\cos(x)dx\right] \quad \text{(Split intervals)} = \frac{1}{\pi}\left[-c\int_{-\pi}^{0} \cos(x)\right.$$

✏ **Result:** $a_1 = 0$ (All a_n coefficients are 0 due to odd symmetry)

Finding b_1:

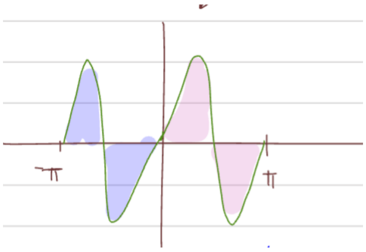
$$b_1 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)\sin(x)dx \qquad = \frac{1}{\pi}\left[\int_{-\pi}^{0} (-c)\sin(x)dx + \int_{0}^{\pi} c\sin(x)dx\right] \quad \text{(Split intervals)} = \frac{1}{\pi}[(-c)[-\cos(x)]^{0}_{-}$$

✏ **Result:** $b_1 = \frac{4c}{\pi}$

Finding b_2:

$$b_2 = \frac{1}{\pi}\left[\int_{-\pi}^{0}(-c)\sin(2x)dx + \int_{0}^{\pi} c\sin(2x)dx\right] \qquad = \frac{1}{\pi}\left[(-c)\left(-\frac{1}{2}\right)[\cos(2x)]_{-\pi}^{0} + c\left(-\frac{1}{2}\right)[\cos(2x)]_{0}^{\pi}\right] \quad \text{(Integ}$$

Result: $b_2 = 0$ (All even harmonics are 0)



Finding b_3:

$$b_3 = \frac{1}{\pi}\left[\int_{-\pi}^{0}(-c)\sin(3x)dx + \int_{0}^{\pi}(c)\sin(3x)dx\right] \qquad = \frac{1}{\pi}\left[(-c)\left(-\frac{1}{3}\right)[\cos(3x)]_{-\pi}^{0} + c\left(-\frac{1}{3}\right)[\cos(3x)]_{0}^{\pi}\right] \quad \text{(Int}$$

Result: $b_3 = \frac{4c}{3\pi}$

Pattern: Even harmonics b_4, b_6, b_8, etc., evaluate to 0.

Resulting Series:

$$f(x) = \frac{4c}{\pi}\sin(x) + \frac{4c}{3\pi}\sin(3x) + \frac{4c}{5\pi}\sin(5x) + \frac{4c}{7\pi}\sin(7x) + \dots$$

Example 2: 8-bit ASCII "b"

Problem: Find the Fourier series representation of the 8-bit ASCII character "b" which translates to the binary signal `01100010`.

Signal Waveform:

```
Bit:    0   1   2   3   4   5   6   7
        0   1   1   0   0   0   1   0


            -------             ---
           |       |           |   |
   1       |       |           |   |
           |       |           |   |
   0_______|       |___________|   |_______
       0   1   2   3   4   5   6   7   8
                           Time →
```

Answer:

Given:

- The period is $P = T = 8$.

Finding a_0:

$$a_0 = \frac{1}{P}\int_0^T g(t)dt \qquad = \frac{1}{8}\left[\int_1^3 1dt + \int_6^7 1dt\right] = \frac{1}{8}(2+1) \qquad = \frac{3}{8}$$

Finding a_n:

$$a_n = \frac{2}{P}\int_0^T g(t)\cos\left(\frac{2\pi nt}{P}\right)dt \qquad = \frac{2}{8}\left[\int_1^3 \cos\left(\frac{\pi nt}{4}\right)dt + \int_6^7 \cos\left(\frac{\pi nt}{4}\right)dt\right] = \frac{1}{4}\cdot\frac{4}{\pi n}\left[\sin\left(\frac{3\pi n}{4}\right) - \sin\left(\right.\right.$$

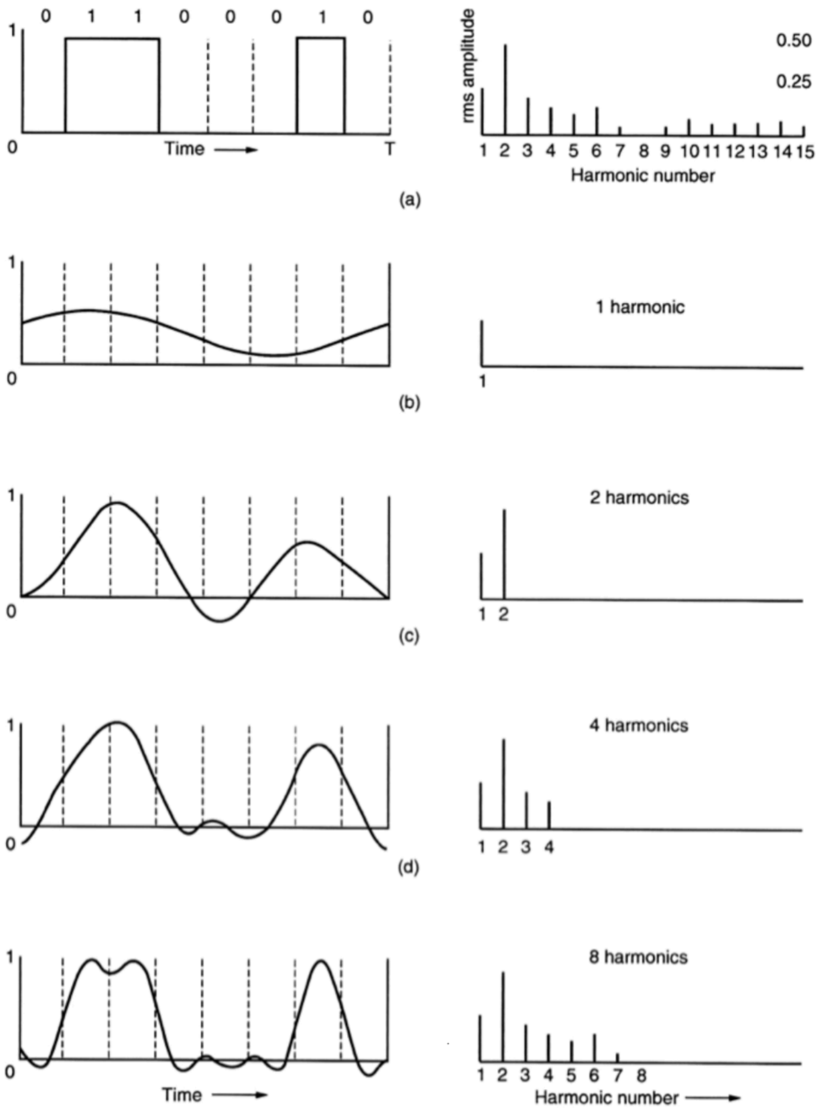
Finding b_n:

$$b_n = \frac{1}{\pi n}\left[\cos\left(\frac{\pi n}{4}\right) - \cos\left(\frac{3\pi n}{4}\right) + \cos\left(\frac{6\pi n}{4}\right) - \cos\left(\frac{7\pi n}{4}\right)\right]$$

RMS & Harmonics

Root Mean Square Amplitude: The value $\sqrt{a_n^2 + b_n^2}$ is important because its square is proportional to the energy transmitted at the corresponding frequency.

Signal Approximation: Adding more harmonics (increasing n) makes the plotted series closer to the original signal.



Signal Transmission Through Channel

```
          Transmitted                  Channel                     Received

              ^                             ^                            ^
              |--------                     |-------                     |--------
              |         |                   |       \                    |        |
              |         |                   |        \                   |        |
--------------+         |---------   ------+         |---------   ------+        |
              |-------> f                   |-------> f                  |-------> f
                     f_s                           f_c                          f_R
```

Two Types of Signals

Baseband Signal

```
Amplitude
    |
    |      _______
    |     |       |
    |_____|       |___________
    |
    +-----|-------|-----------> Frequency (Hz)
          0      fs
```

Signal transmitted at original frequency, starting from 0 Hz

Passband Signal

```
Amplitude
    |
    |            _______
    |           |       |
    |___________|       |_____
    |
    +-----|-----|-------|----> Frequency (Hz)
          0    fa       fb
```

Signal shifted to higher frequency band for transmission

Data Rate and Channel Harmonics

Signals can generally be classified as either baseband signals or passband signals.

Problem: Given a data rate of b bits/sec, how many harmonics can be transmitted through a voice-grade line with bandwidth 3000 Hz?

Answer:

Step 1: Calculate the time to send 8 bits:

$$T = \frac{8}{b} \text{ seconds}$$

Step 2: Calculate the frequency of the first harmonic:

$$f_1 = \frac{1}{T} = \frac{b}{8} \text{ Hz}$$

Step 3: Calculate the maximum number of harmonics in a 3 kHz bandwidth:

$$\text{Number of harmonics} = \frac{3000}{f_1} = \frac{3000}{b/8} = \frac{24000}{b}$$

Relation Between Data Rate and Harmonics

How is data rate related to harmonics?

Given a data rate of b bits/sec, the time required to send 8 bits is:

$$T = \frac{8}{b} \text{ seconds}$$

The frequency of the first harmonic (minimum) is:

$$f_1 = \frac{1}{T} = \frac{b}{8} \text{ Hz}$$

A voice-grade line has a bandwidth of 3000 Hz (3 kHz). This means that the largest number of harmonics that can be sent through is:

$$\text{Max harmonics} = \frac{3000}{f_1} = \frac{3000}{b/8} = \frac{24000}{b}$$

Example Calculation (300 bps):

$$T = \frac{8}{300} = 26.67 \text{ msec} f_1 \qquad = \frac{300}{8} = 37.5 \text{ Hz} \text{Max harmonics} = \frac{3000}{37.5} = 80$$

Bps	T (msec)	First harmonic (Hz)	# Harmonics sent
300	26.67	37.5	80
600	13.33	75	40
1200	6.67	150	20
2400	3.33	300	10

Bps	T (msec)	First harmonic (Hz)	# Harmonics sent
4800	1.67	600	5
9600	0.83	1200	2
19200	0.42	2400	1
38400	0.21	4800	0

Channel Capacity Theorems

Nyquist Capacity (1924, AT&T)

Defines the maximum data rate for a noiseless channel.

Formula:

$$\text{Maximum data rate} = 2B \log_2(V) \text{ bits/sec}$$

Variables:

- B = bandwidth in Hz
- V = number of discrete levels in the signal

Problem: What is the maximum data rate over a noiseless channel with a 3 kHz bandwidth using a binary (two levels) signal?

Answer:

$$\text{Maximum data rate} = 2B \log_2(V) \qquad = 2 \times 3000 \times \log_2(2) = 6000 \text{ bps}$$

Shannon Capacity (1948)

Defines the maximum data rate for a noisy channel.

Formula:

$$\text{Maximum data rate} = B \log_2(1 + \text{SNR}) \text{ bits/sec}$$

Signal-to-Noise Ratio (SNR):

- SNR is measured in decibels (dB)
- $\text{SNR}_{\text{dB}} = 10 \log_{10}(\text{SNR})$
- **Example:** An SNR of 1000 equates to $10 \log_{10}(1000) = 30$ dB

Problem: What is the line capacity of an ADSL line with a bandwidth of 1 MHz and an SNR of 40 dB?

Answer:

Step 1: Convert SNR from dB to linear scale:

$$40 = 10 \log_{10}(\text{SNR})\text{SNR} \qquad = 10^4 = 10000$$

Step 2: Calculate line capacity:

$$C = B \log_2(1 + \text{SNR}) \qquad = 10^6 \times \log_2(1 + 10000) = 10^6 \times \log_2(10001) \qquad \approx 10^6 \times 13.29 \approx 13 \text{ Mbps}$$